Module 16: Capstone Systems Synthesis - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. A strong biological explanation that spans scales does what?
2. A. stays at a single scale only
3. B. avoids using any evidence
4. C. connects mechanisms from molecules up to ecosystems into one account
5. D. relies on vocabulary alone
6. Answer: C

7. Why: Synthesis links causes across scales, from molecular to ecological, into one coherent, evidence-backed explanation.

8. When rising body temperature makes you sweat and cool back down, that is an example of:

9. A. a negative feedback loop that reduces change
10. B. a random mutation
11. C. a tradeoff with no feedback
12. D. genetic drift
13. Answer: A

14. Why: Negative feedback counteracts a change to keep conditions stable, which is a core systems idea behind homeostasis.

15. "A larger body helps an animal retain heat but also requires more food." This statement describes a:

16. A. feedback loop
17. B. mutation
18. C. species concept
19. D. tradeoff

20. Answer: D
21. Why: A tradeoff pairs a benefit in one function with a cost in another, a common pattern in biological systems.
22. To support a claim that spans several biological scales, the strongest evidence is:
23. A. a single memorable anecdote
24. B. an evidence chain in which each step's evidence matches its scale
25. C. a list of vocabulary words
26. D. personal opinion alone
27. Answer: B
28. Why: An evidence chain links the claim to appropriate data at each scale, which makes a cross-scale explanation testable and convincing.